

PRST Class (22/09/2020).

Continuous rv — It takes any value within a certain range.

~~X~~ X is a cont rv $(-\infty < x < \infty)$.

$$P(a \leq X \leq b) = \int_a^b f(x) dx.$$

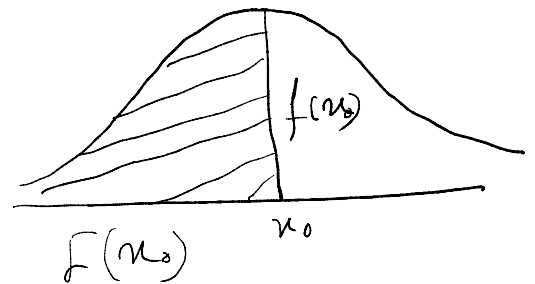
$f(x)$ — probability density function (pdf).

$$F(x_0) = P(X \leq x_0) = \int_{-\infty}^{x_0} f(x) dx.$$

— The funcn $f(x)$ is said to be a pdf if

(i) $f(x) \geq 0$

(ii) $\int_{-\infty}^{\infty} f(x) dx = 1.$



$F(x)$ — distribution function
 $F(x) = P(X \leq x)$

(i) $F(x)$ is increasing. $\left| \begin{array}{l} x_1 < x_2 \\ F(x_1) \leq F(x_2) \end{array} \right.$

(ii) $F(\infty) = 1$

(iii) $F(-\infty) = 0$

$\lim_{x_0 \rightarrow \infty} F(x_0)$

~~(iv)~~

X is cont rv.

$P(X = x_0) = P(x_0 \leq X \leq x_0)$

$\int_{x_0}^{x_0} f(x) dx = 0$

→

Eg $0 \leq X \leq 1 \rightarrow$
 $f(x) = 1 \quad 0 \leq x \leq 1$
~~P~~ $F(0) = 0$
 $F(1) = 1$

$$\int_{x_0}^{x_1} f(x) dx = 0$$

~~165 cm~~
~~165.01 cm~~
~~165.008 cm~~

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

$$P(a < X \leq b) = \int_a^b f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

E_X

X	$P(X=x)$
1	k
2	$2k$
3	$2k$
4	$3k$
5	k^2
6	$2k^2$
7	$7k^2 + k$

(i) find the value of k .

(ii) find $F(6)$.

$$(k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k) = 1$$

$$k = -1, \frac{1}{10}$$

$$\therefore k = \frac{1}{10}$$

$$k \geq 0 \Rightarrow k = \frac{1}{10}$$

$$F(6) = 0.84 \cdot 0.83$$

E_x

$$f(x) = \begin{cases} 2x & 0 < x \leq 1 \\ 4-2x & 1 < x \leq 2 \\ 0 & \text{o.w.} \end{cases}$$

$f(x)$ is a pdf?

(i) $f(x) \geq 0$ ✓

(ii) $\int_{-\infty}^{\infty} f(x) dx = \int_0^2 f(x) dx = \int_0^1 2x dx + \int_1^2 (4-2x) dx$

$$= \frac{2}{2} + 1 = 2 > 1$$

$\therefore f(x)$ is not a pdf.

E_x

X is a continuous rv, $f(x) = \begin{cases} \frac{1}{4} & 2 \leq x \leq 2 \\ 0 & \text{o.w.} \end{cases}$

$P(2X+3 > 5) = ?$

$P(X > 1) = \int_1^2 f(x) dx = \int_1^2 \frac{1}{4} dx$

$$= \frac{1}{4}$$

cont

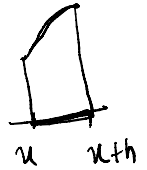
$F(x) \xrightarrow{?} f(x)$

$f(x) = \frac{dF(x)}{dx}$

$\square(x)$

$$f(x) = \frac{dF(x)}{dx}$$

$$\begin{aligned} f(x) &= \lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{P(X \leq x+h) - P(X \leq x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{P(x \leq X < x+h)}{h} \end{aligned}$$



$$\Omega = \{HHHH, THH, TTH, \dots\}$$

$X_1 = \# \text{ heads in outcome}$

$X_2 = \# \text{ tails in outcome.}$

$$(X_1, X_2)$$